Programming 2

SEMM Primer 2026
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Outline

- Review of loops
- Review of methods/functions
- Example 1: File input/output
- Example 2: Newton Raphson
- OpenSees intro
- Walkthrough of OpenSees code

Hands on in Jupyter Lab / Colab

Programming basics - Loops

- For loop
  - You know how many times to loop
  - `for i in range(5):`
  - `for n in [1, 2, 3, 4]:`
  - `For i, letter in enumerate([‘a’, ‘b’, ‘c’]):`
  - Can stop early if you want
- While loop
  - You don’t know how many times to loop
  - Need to always be able to stop the loop
    - Condition to loop (`while error < tolerance:`)
    - `break`

Programming basics - Methods/functions

- Do not copy paste lines of code, put it into a method/function!
- If you use the same chunk of code multiple times, try to put it into a function
- `def increment_all(values):`
  - `answer = []`
  - `for v in values:`
    - `answer.append(v + 1)`
  - `return answer`
- `def increment_all(values):`
  - `return [i + 1 for i in values]` (w/ list comprehension)

Programming basics - Methods/functions

- Functions can return things or edit the input
- `def increment_all(values):`
  - `for i in range(len(values)):`
    - `values[i] = values[i] + 1`
- Function parameters can have default values
  - `def integrate(values, dx=0.1):`
- You can decorate function parameters for readability
  - `def integrate(values:”list of values”, dx:”step”=0.1) -> “trapezoidal rule sum”:`

Programming basics - Methods/functions

Variables inside the function are locally scoped to that function

```
A = 10

Def foo():

    A = 5

    return A

print(A) # This will print 10

print(foo()) # This will print 5
```

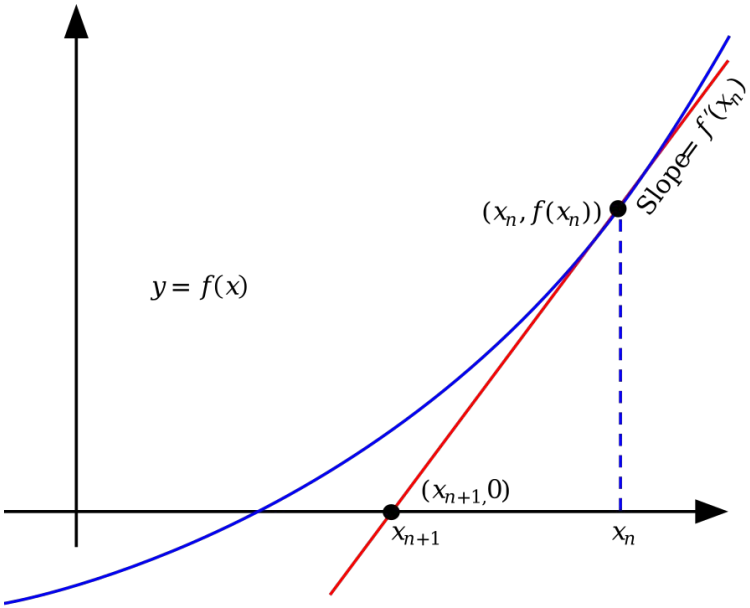
Example 1: File input / output

- Read in ground motion data from NGA West 2
  - https://peer.berkeley.edu/research/nga-west-2
- Plot it
- Integrate it twice to get displacements
- Rotate the displacement orbits

Example 2: Newton Raphson

- Solve for the root of a function
- Iterative approach
- Just needs the derivative
- Useful for vector calculations (`f = Ku`)

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

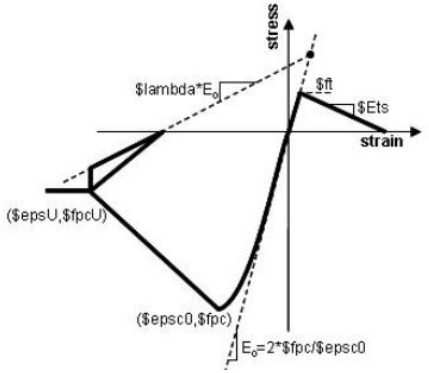


- Need derivative to not be 0 (i.e. not singular)

OpenSees

- https://opensees.berkeley.edu/
- Structural analysis software using finite element approach
- Common in research
- More efficient, transparent, editable, adaptable than SAP2000/ETABS
  - But, much higher learning curve, easier to make mistakes
  - Need to develop own plotting and visualization to see outputs
  - Need to be consistent with units, coordinate systems, dof numbering, local degrees of freedom
  - Need to know what you're doing! Check with hand calcs! **Does it make sense?**
- Can work with it in Python (OpenSeesPy, Xara)

OpenSees - Main workflow



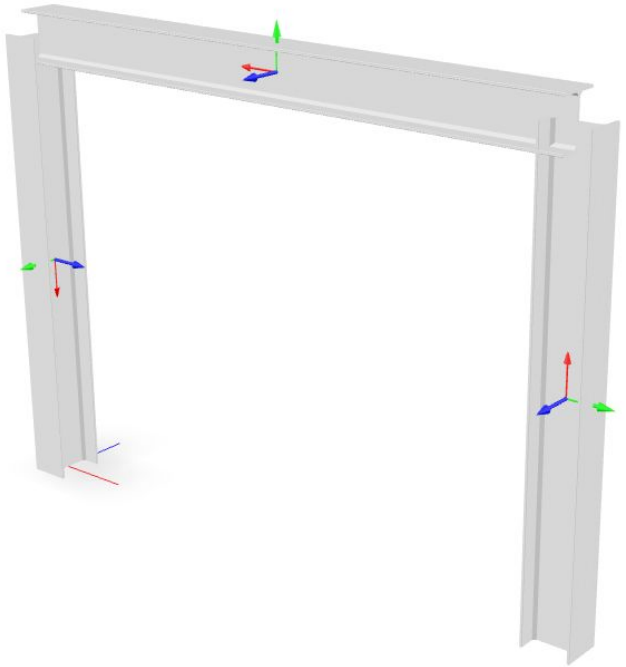
https://opensees.berkeley.edu/wiki/index.php/Concrete02_Material_--_Linear_Tension_Softening

Define

- Material responses (concrete, steel)
- Element types, sections and orientations
  - E.x. fiber sections, linear elastic prisms, zeroLength springs
- Geometry (nodes, element connections, fixed dofs)
- Loads (masses, load directions and nodes, uniform accelerations)
- Analysis (analysis settings, loop)
- Recorders (output nodal deformations, reactions, element stresses, strains)
- Post processing (read recorder outputs and make sense of it all)

OpenSees - Coordinate system

- Define 2D or 3D system
  - 3 dofs in 2D
  - 6 dofs in 3D
- Orientation of sections matter (geomTransf)
- Choose your global coordinate system
  - Usually ‘Y’ or ‘Z’ axis is vertically upwards
  - Follow right-hand rule conventions
- Local coordinate system
  - Each element has local ‘x’ from node i to node j
  - Local ‘y’ and ‘z’ are based on the geomTransf used
  - Available output local strains are based on element
    - E.x. beam column has axial strain, 2 curvatures at each end, and torsional dof



https://xara.so/examples/frames/frame-0059.html

OpenSees - Quirks

- Everything has a tag
  - Material tag, element tag, node tag, load pattern tag, time history tag....
- Element node order matters
  - Node 1 -> Node 2 is not the same as Node 2 -> Node 1
  - https://openseesdigital.com/2022/05/08/switching-sides/
- Recorder outputs are not labelled
  - Order of outputs may be different, usually axial, two shears, torsion, two moments
  - Need to do some detective work to properly label
- Error outputs are not obvious
  - Write own debug statements / logging errors
  - Start from simple examples and build up piece by piece

Walkthrough 1: Uniaxial materials

https://xara.so/examples/material/material-0002.html

https://opensees.berkeley.edu/wiki/index.php/UniaxialMaterial_Command

Walkthrough 2: OpenSees Moment Curvature Analysis

https://xara.so/examples/sections/fiber-0004.html

https://openseespydoc.readthedocs.io/en/latest/src/MomentCurvature.html